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Narula Institute of Technology
An Autonomous Institute under MAKAUT
2024
END SEMESTER EXAMINATION - ODD 2024
PH(IT)101 - ENGINEERING PHYSICS - I

TIME ALLOTTED: 3Hours**FULL MARKS: 70**

Instructions to the candidate:

Figures to the right indicate full marks.

Draw neat sketches and diagram wherever is necessary.

Candidates are required to give their answers in their own words as far as practicable

Group A

(Multiple Choice Type Questions)

Answer any ten from the following, choosing the correct alternative of each question: 10×1=10

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|--|-----|-----|-----|
| 1.i) What is called coherence? | (1) | CO1 | BL1 |
| 1.ii) Write the expression of Numerical Aperture. | (1) | CO1 | BL1 |
| 1.iii) In Holography ,.....image is formed. (a) 0D, (b) 1D (c) 2D (d) 3D | (1) | CO1 | BL1 |
| 1.iv) What happens to the resistance of a pure semiconductor when heated? (a)The resistance increases, (b) The resistance decreases (c) The temperature remains the same (d) Can't say | (1) | CO2 | BL2 |
| 1.v) The concept of matter wave was suggested by (a) Heisenberg (b) De Broglie (c) Schrodinger (d) Laplace | (1) | CO5 | BL1 |
| 1.vi) The wavelength of de Broglie waves is independent of: (a) charge (b) momentum (c) velocity (d) mass | (1) | CO5 | BL1 |
| 1.vii) Amorphous solids have _____ structure.
(a) Regular
(b) Linear
(c) Irregular
(d) Dendritic | (1) | CO3 | BL1 |
| 1.viii) What is the full form of CNT? (a) carbon nanotube (b) car nan-tube (c) carbon nylon tube (d) carbon micro-tube | (1) | CO2 | BL1 |
| 1.ix) Minimum no. of energy levels required for sustainable laser action is
a) 1
b) 2
c) 3
d) 4 | (1) | CO1 | BL1 |
| 1.x) Numerical Aperture of optical fibre represents | (1) | CO1 | BL2 |

- a) light transmission power
- b) light collecting power
- c) light absorbing power
- d) light scattering power

- 1.xi) The basic physical phenomenon involved in optical fiber communication is (1) CO1 BL2
- a) refraction
 - b) interference
 - c) polarization
 - d) total internal reflection
- 1.xii) The CRO is used to display a that varies as a function of time. (1) CO4 BL1
- (a) waveform
 - (b) wave-length
 - (c) wave-particle duality
 - (d) None of the above

Group B

(Short Answer Type Questions)

(Answer any three of the following) 3x5=15

2. Explain the relationship between numerical aperture and acceptance angle in optical fibers. The acceptance angle of a given optical fiber is 45 degree. Calculate the numerical aperture of that optical fiber. (2.5+2.5=5) (5) CO1 BL1
3. What is Miller Indices? Given the intercepts of a plane as 1, 2, and 3 along the x, y, and z axes respectively, calculate the Miller indices. (5) CO3 BL2
4. State Bragg's Law for x-ray diffraction. A crystal shows a first-order diffraction peak at an angle of 30° for X-rays of wavelength 1.54 Å. Calculate the interplanar spacing. (2+3=5) (5) CO3 BL1
5. Write the expression of Compton Wave-length. Calculate the Compton wave length of the electron and proton. (1+2+2=5) (5) CO5 BL2
- 6a. Draw the energy distribution of the spectrum of a black body. How does the spectrum vary with temperature changes? Explain. (3) CO3 BL3
- 6b. What are the advantages of optical storage? (2) CO3 BL4

Group C

(Long Answer Type Questions)

(Answer any three of the following) 3x15=45

7. (a) What is LASER? Write the basic properties of LASER. (1+2) (15) CO1 BL3
- (b) Derive the relation between Einstein's A and B Coefficients. (6)
- (c) What is Population Inversion? (3)

(d) What is a Meta-stable state? (2)

(e) What is the value of LASER wavelength in Ruby LASER? (1)

8. (a) Calculate the atomic packing factor of BCC lattice. (15) CO3 BL3

(b) Show that the distance between two successive plane in simple cubic crystal is

$$d = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

(c) For a cubic crystal, calculate the interplanar spacing d for the (1, 1, 1) plane if the lattice parameter a is 4 Å.

(d) State the Bragg's Law? Write its Significance.

(e) The 1st order reflection from the plane of NaCl is obtained at an angle 30° . If $d = 2.82 \text{ Å}$ then calculate the wave-length of x-ray used.

(3+3+3+3+3=15)

9. (a) Define the coordination number. Write the coordination number for atoms in a SC and BCC lattice. (1+1+1) (15) CO3 BL3

(b) What is the atomic packing factor (APF), and how is it calculated for a simple cubic structure? How it is related to void space?

(1+3+1=5)

(c) State the **Bragg's law**. Explain the significance of Bragg's law in X-ray crystallography. (2+2=4)

(d) A crystal with interplanar spacing of $d = 2.5 \text{ Å}$ produces a diffraction peak at an angle of $\theta = 30^\circ$. Calculate the wavelength of the X-rays used. (3)

10a. Explain the operation of a He Ne LASER with a suitable diagram. (5) CO2 BL3

10b. What is the difference between a step index and a graded index fiber. (5) CO2 BL3

10c. Calculate the atomic packing fraction and atoms per unit cell in crystals having Simple cubic, body centered cubic and face centered cubic structure considering the atoms as hard sphere. (5) CO2 BL3

11a. What is Miller indices? (2) CO1 BL2

11b. Find the Miller indices of a plane which intercepts at a , $b/2$, $3c$ in simple cubic unit cell. (4) CO2 BL3

11c. What are the different losses in an fiber? The attenuation of an optical fiber is 2.6 dB/km. If the power of optical signal at launching end is 0.6mW, find the power level after 5km. (5) CO2 BL3

11d. The distance between (1,1,0) plane in FCC structure is 0.246nm. what is the size of unit cell? What is the radius of the atom? (4) CO2 BL3